

Passaggio parametri versione n. 1

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([IN OUT x rif] int r)
    int s; int t;
    procedure p3([MODE] int s,[IN OUT x copia] int q)
      int t;
      BEGIN
        t=s-1;
        if t=8 then s=q else s=2;
        q=t;
        r=r-4;
        write(q,r,s,t);
      END

      BEGIN
        s=q;
        t=q+4;
        r=q-2;
        q=r;
        p3(r, t);
        write(q,r,s,t);
      END

    procedure p4([IN OUT x copia] int r)
      int t; int s;
      BEGIN
        t=1;
        s=1;
        r=2;
        q=t-2;
        p2(q);
        write(q,r,s,t);
      END

    BEGIN
      q=3;
      r=3;
      s=1;
      t=0;
      p4(r);
      write(q,r,s,t);
    END
```

Passaggio parametri versione n. 2

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN OUT per riferimento

```
program p1
int a; int b; int c; int d;
  procedure p2([MODE] int d)
    int b; int c;
    BEGIN
      b=3;
      if a<1 then c=2 else c=4;
      d=d;
      a=a;
      p4(a, c);
      write(a,b,c,d);
    END

    procedure p3([IN x copia] int c,[IN x copia] int b)
      int a;
      BEGIN
        a=3;
        if c=3 then c=c else c=d+4;
        b=3;
        d=c+4;
        write(a,b,c,d);
      END

      procedure p4([IN x copia] int d,[IN OUT x copia] int b)
        int a;
        BEGIN
          a=2;
          d=b;
          b=1;
          c=b;
          p3(b, a);
          write(a,b,c,d);
        END

      BEGIN
        a=4;
        b=2;
        c=3;
        d=0;
        p2(d);
        write(a,b,c,d);
      END
```

Passaggio parametri versione n. 3

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([MODE] int q,[IN x copia] int s)
    int r;
      procedure p3([IN OUT x rif] int q,[IN OUT x copia] int s)
        int t;
        BEGIN
          if r>0 then t=1 else t=2;
          q=4;
          s=r;
          r=t;
          p4(r);
          write(q,r,s,t);
        END

      BEGIN
        if q<50 then r=1 else r=t;
        q=t+3;
        s=2;
        t=s-2;
        p3(s, t);
        write(q,r,s,t);
      END

    procedure p4([IN OUT x copia] int s)
      int t;
      BEGIN
        t=1;
        s=s*1;
        if s=20 then q=1 else q=s;
        r=q+2;
        write(q,r,s,t);
      END

    BEGIN
      q=1;
      r=1;
      s=2;
      t=2;
      p2(r, s);
      write(q,r,s,t);
    END
```

Passaggio parametri versione n. 4

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per riferimento

```
program p1
int x; int y; int z; int t;
  procedure p2([MODE] int t)
    int y; int x;
    BEGIN
      y=2;
      x=2;
      t=3;
      z=1;
      p3(x);
      write(x,y,z,t);
    END

    procedure p3([IN x copia] int x)
      int z;
      BEGIN
        z=1;
        x=t*2;
        t=z;
        y=2;
        write(x,y,z,t);
      END

      procedure p4([IN x copia] int t,[IN OUT x copia] int y)
        int z;
        BEGIN
          z=x;
          t=2;
          y=z;
          x=x*1;
          p2(t);
          write(x,y,z,t);
        END

      BEGIN
        x=2;
        y=1;
        z=2;
        t=1;
        p4(y, x);
        write(x,y,z,t);
      END
```

Passaggio parametri versione n. 5

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN OUT per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([MODE] int s)
    int r; int t;
    BEGIN
      r=s-1;
      t=2;
      s=q;
      q=t+4;
      write(q,r,s,t);
    END

    procedure p3([IN x copia] int s,[IN x copia] int r)
      int q;
      BEGIN
        q=r;
        s=4;
        r=3;
        t=1;
        p2(q);
        write(q,r,s,t);
      END

      procedure p4([IN OUT x rif] int s)
        int t;
        BEGIN
          t=q+2;
          s=s;
          if t>50 then q=4 else q=3;
          r=t;
          p3(q, t);
          write(q,r,s,t);
        END

      BEGIN
        q=0;
        r=4;
        s=1;
        t=1;
        p4(t);
        write(q,r,s,t);
      END
```

Passaggio parametri versione n. 6

- Scoping dinamico, MODE = IN per copia
- Scoping statico, MODE = OUT per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([MODE] int q)
    int s;
    procedure p3([IN OUT x rif] int q,[IN OUT x rif] int t)
      int r;
      procedure p4([IN OUT x copia] int r)
        int s; int q;
        BEGIN
          s=4;
          q=t+2;
          r=4;
          t=t+2;
          write(q,r,s,t);
        END

      BEGIN
        r=s+2;
        q=1;
        t=t-3;
        s=1;
        p4(q);
        write(q,r,s,t);
      END

    BEGIN
      s=r;
      r=q;
      t=3;
      p3(s, t);
      write(q,r,s,t);
    END

  BEGIN
    q=1;
    r=4;
    s=3;
    t=2;
    p2(q);
    write(q,r,s,t);
  END
```

Passaggio parametri versione n. 7

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN OUT per riferimento

```
program p1
int q; int r; int s; int t;
  procedure p2([MODE] int t)
    int s;
    BEGIN
      s=t*1;
      t=4;
      q=3;
      r=r;
      p3(t);
      write(q,r,s,t);
    END

    procedure p3([IN x copia] int r)
      int t;
      procedure p4([IN x copia] int t)
        int q; int s;
        BEGIN
          q=t;
          s=t*4;
          t=q;
          r=s*4;
          write(q,r,s,t);
        END

        BEGIN
          t=s;
          r=4;
          if t<3 then q=3 else q=t+3;
          s=r;
          p4(q);
          write(q,r,s,t);
        END

      BEGIN
        q=1;
        r=1;
        s=2;
        t=3;
        p2(r);
        write(q,r,s,t);
      END
```

Passaggio parametri versione n. 8

- Scoping dinamico, MODE = OUT per riferimento
- Scoping statico, MODE = IN OUT per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([IN x copia] int q)
    int s; int r;
    BEGIN
      s=q*2;
      r=t;
      q=t+4;
      t=3;
      p3(t, s);
      write(q,r,s,t);
    END

    procedure p3([MODE] int r,[IN x copia] int q)
      int t;
      procedure p4([IN x copia] int t)
        int s; int r;
        BEGIN
          if t<7 then s=1 else s=q;
          r=t*2;
          t=r;
          q=1;
          write(q,r,s,t);
        END

        BEGIN
          t=r;
          if q=2 then r=1 else r=t;
          q=q+1;
          s=q;
          p4(t);
          write(q,r,s,t);
        END

      BEGIN
        q=4;
        r=3;
        s=3;
        t=4;
        p2(q);
        write(q,r,s,t);
      END
```


Passaggio parametri versione n. 9

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN per copia

```
program p1
int a; int b; int c; int d;
  procedure p2([IN OUT x rif] int c)
    int d;
    BEGIN
      d=b;
      if b=10 then c=a+2 else c=1;
      a=c*4;
      if a=50 then b=b*3 else b=3;
      write(a,b,c,d);
    END

    procedure p3([MODE] int b)
      int d; int c;
      procedure p4([IN OUT x copia] int a)
        int c;
        BEGIN
          c=d;
          a=d+4;
          b=3;
          if a=20 then d=b else d=2;
          p2(c);
          write(a,b,c,d);
        END

        BEGIN
          d=b;
          if a>1 then c=3 else c=a;
          b=1;
          a=c;
          p4(c);
          write(a,b,c,d);
        END

      BEGIN
        a=3;
        b=1;
        c=3;
        d=2;
        p3(c);
        write(a,b,c,d);
      END
```

Passaggio parametri versione n. 10

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN per riferimento

```
program p1
int a; int b; int c; int d;
  procedure p2([MODE] int b)
    int a;
      procedure p3([IN OUT x rif] int b)
        int c; int a;
        BEGIN
          c=d+3;
          a=1;
          b=a;
          d=a;
          p4(d);
          write(a,b,c,d);
        END

      BEGIN
        a=c+4;
        b=1;
        c=3;
        if a>6 then d=4 else d=d;
        p3(a);
        write(a,b,c,d);
      END

    procedure p4([IN OUT x rif] int a)
      int b;
      BEGIN
        b=3;
        a=a-4;
        d=2;
        c=2;
        write(a,b,c,d);
      END

    BEGIN
      a=0;
      b=1;
      c=2;
      d=4;
      p2(d);
      write(a,b,c,d);
    END
```

Passaggio parametri versione n. 11

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = IN OUT per copia

```
program p1
int x; int y; int z; int t;
  procedure p2([IN OUT x rif] int z)
    int y; int x;
    BEGIN
      y=z+1;
      x=3;
      z=4;
      t=t*2;
      write(x,y,z,t);
    END

    procedure p3([MODE] int x)
      int y;
      procedure p4([IN x copia] int y)
        int z; int x;
        BEGIN
          z=4;
          x=3;
          if t>20 then y=y else y=t+3;
          t=t*2;
          p2(x);
          write(x,y,z,t);
        END

        BEGIN
          y=t*2;
          x=x*3;
          z=4;
          t=4;
          p4(x);
          write(x,y,z,t);
        END

      BEGIN
        x=3;
        y=4;
        z=1;
        t=0;
        p3(x);
        write(x,y,z,t);
      END
```

Passaggio parametri versione n. 12

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int a; int b; int c; int d;
  procedure p2([IN OUT x rif] int d,[MODE] int b)
    int c;
    BEGIN
      c=a;
      d=a+1;
      if d>5 then b=a-3 else b=c;
      a=4;
      p4(b);
      write(a,b,c,d);
    END

    procedure p3([IN x copia] int c)
      int b;
      BEGIN
        if a=4 then b=1 else b=a;
        c=1;
        d=c;
        a=1;
        p2(b, d);
        write(a,b,c,d);
      END

      procedure p4([IN OUT x rif] int c)
        int b; int d;
        BEGIN
          b=c;
          d=2;
          c=d+3;
          a=2;
          write(a,b,c,d);
        END

      BEGIN
        a=0;
        b=3;
        c=3;
        d=4;
        p3(b);
        write(a,b,c,d);
      END
```

Passaggio parametri versione n. 13

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN OUT per copia

```
program p1
int a; int b; int c; int d;
  procedure p2([MODE] int b)
    int d;
    BEGIN
      d=b;
      b=3;
      a=d-1;
      c=4;
      p3(b, c);
      write(a,b,c,d);
    END

    procedure p3([IN OUT x rif] int d,[IN OUT x rif] int c)
      int a;
      procedure p4([IN x copia] int b)
        int d;
        BEGIN
          d=a+3;
          b=c;
          a=4;
          c=c+2;
          write(a,b,c,d);
        END

        BEGIN
          if d=8 then a=3 else a=4;
          if b<8 then d=a else d=b+1;
          c=d*2;
          b=1;
          p4(c);
          write(a,b,c,d);
        END

      BEGIN
        a=1;
        b=1;
        c=4;
        d=3;
        p2(b);
        write(a,b,c,d);
      END
```

Passaggio parametri versione n. 14

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN OUT per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([MODE] int q)
    int t; int s;
      procedure p3([IN x copia] int t)
        int r;
        BEGIN
          r=1;
          t=q;
          q=t-1;
          s=1;
          write(q,r,s,t);
        END

      BEGIN
        t=q;
        if q=4 then s=q+4 else s=q;
        q=1;
        r=s-1;
        p3(t);
        write(q,r,s,t);
      END

    procedure p4([IN OUT x rif] int r)
      int q;
      BEGIN
        q=s-1;
        r=r*4;
        if s=1 then s=q-4 else s=4;
        t=2;
        p2(r);
        write(q,r,s,t);
      END

    BEGIN
      q=0;
      r=3;
      s=2;
      t=1;
      p4(t);
      write(q,r,s,t);
    END
```

Passaggio parametri versione n. 15

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = IN OUT per copia

```
program p1
int x; int y; int z; int t;
  procedure p2([MODE] int z)
    int x;
      procedure p3([IN x copia] int x,[IN OUT x rif] int y)
        int t;
          procedure p4([IN x copia] int y)
            int t; int z;
            BEGIN
              t=3;
              if x<20 then z=2 else z=x;
              y=3;
              x=2;
              write(x,y,z,t);
            END

            BEGIN
              t=1;
              x=4;
              y=3;
              z=2;
              p4(y);
              write(x,y,z,t);
            END

            BEGIN
              x=2;
              if x<9 then z=t*1 else z=y;
              y=z*1;
              if x=3 then t=2 else t=z;
              p3(x, z);
              write(x,y,z,t);
            END

            BEGIN
              x=1;
              y=0;
              z=4;
              t=2;
              p2(x);
              write(x,y,z,t);
            END
```

Passaggio parametri versione n. 16

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int q; int r; int s; int t;
procedure p2([IN OUT x copia] int s)
  int r;
    procedure p3([IN OUT x rif] int s)
      int q; int t;
      BEGIN
        q=r;
        t=2;
        s=4;
        r=3;
        p4(q);
        write(q,r,s,t);
      END

    procedure p4([MODE] int r)
      int s; int q;
      BEGIN
        s=3;
        q=t;
        r=1;
        t=2;
        write(q,r,s,t);
      END

    BEGIN
      if s<20 then r=2 else r=s-3;
      s=r;
      q=t*2;
      t=1;
      p3(s);
      write(q,r,s,t);
    END

  BEGIN
    q=1;
    r=4;
    s=3;
    t=1;
    p2(q);
    write(q,r,s,t);
  END
```


Passaggio parametri versione n. 17

- Scoping dinamico, MODE = IN OUT per riferimento
- Scoping statico, MODE = IN per riferimento

```
program p1
int x; int y; int z; int t;
  procedure p2([IN OUT x copia] int z)
    int t;
    procedure p3([MODE] int z)
      int t;
      procedure p4([IN x copia] int z)
        int y;
        BEGIN
          y=1;
          z=z*4;
          t=3;
          x=t*1;
          write(x,y,z,t);
        END

      BEGIN
        t=4;
        if t<6 then z=y-2 else z=4;
        y=2;
        x=z;
        p4(z);
        write(x,y,z,t);
      END

    BEGIN
      t=y-4;
      z=4;
      y=x;
      x=1;
      p3(t);
      write(x,y,z,t);
    END

  BEGIN
    x=1;
    y=4;
    z=2;
    t=1;
    p2(y);
    write(x,y,z,t);
  END
```

Passaggio parametri versione n. 18

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN per riferimento

```
program p1
int x; int y; int z; int t;
  procedure p2([IN OUT x rif] int t)
    int x; int y;
      procedure p3([IN OUT x rif] int t,[IN OUT x rif] int z)
        int y;
          procedure p4([MODE] int t,[IN OUT x copia] int x)
            int z;
            BEGIN
              z=1;
              t=x;
              x=z;
              y=y;
              write(x,y,z,t);
            END

            BEGIN
              y=x;
              t=y;
              z=t*1;
              x=t;
              p4(z, x);
              write(x,y,z,t);
            END

            BEGIN
              x=t+1;
              y=z;
              t=y*4;
              z=x;
              p3(y, z);
              write(x,y,z,t);
            END

            BEGIN
              x=0;
              y=0;
              z=3;
              t=2;
              p2(x);
              write(x,y,z,t);
            END
```

Passaggio parametri versione n. 19

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([IN OUT x copia] int s,[MODE] int r)
    int q;
      procedure p3([IN OUT x rif] int t,[IN x copia] int s)
        int r;
        BEGIN
          r=t-4;
          t=s*1;
          s=4;
          q=r+1;
          p4(s);
          write(q,r,s,t);
        END

      BEGIN
        if r=1 then q=s else q=t;
        s=4;
        if q=10 then r=t else r=t*2;
        t=1;
        p3(r, s);
        write(q,r,s,t);
      END

    procedure p4([IN OUT x copia] int q)
      int r; int t;
      BEGIN
        r=2;
        t=3;
        q=t;
        s=t*1;
        write(q,r,s,t);
      END

    BEGIN
      q=4;
      r=3;
      s=0;
      t=3;
      p2(s, r);
      write(q,r,s,t);
    END
```